RAMAVANA

MATHEMATICAL CENTRE · HSR Layout, Bengaluru

Straight Lines

Grade 11 · CBSE · Practice worksheet

Name: Date: Batch:

Solve every question in your notebook and show all working. Questions follow the current CBSE syllabus (2026–27). Normal form, general equation, family of lines and shifting of origin are not in the syllabus, so they are not asked here.

Key formulas

Slope through (x_1, y_1) and (x_2, y_2)	$m = \dfrac{y_2 - y_1}{x_2 - x_1},\ x_1 \neq x_2$
Slope from inclination θ	$m = \tan\theta,\ \theta \neq 90^\circ$
Parallel lines	$m_1 = m_2$
Perpendicular lines	$m_1 m_2 = -1$
Acute angle between two lines	$\tan\theta = \left\lvert \dfrac{m_2 - m_1}{1 + m_1 m_2} \right\rvert$
Line parallel to the x-axis / y-axis	$y = b \quad / \quad x = a$
Point–slope form	$y - y_1 = m(x - x_1)$
Two-point form	$y - y_1 = \dfrac{y_2 - y_1}{x_2 - x_1}(x - x_1)$
Slope–intercept form	$y = mx + c$
Intercept form	$\dfrac{x}{a} + \dfrac{y}{b} = 1$
Slope of $Ax + By + C = 0$	$m = -\dfrac{A}{B}$
Distance of (x_1, y_1) from $Ax + By + C = 0$	$d = \dfrac{\lvert Ax_1 + By_1 + C \rvert}{\sqrt{A^2 + B^2}}$
Distance between $Ax + By + C_1 = 0$ and $Ax + By + C_2 = 0$	$d = \dfrac{\lvert C_1 - C_2 \rvert}{\sqrt{A^2 + B^2}}$

Section A — Very short answer

1 mark each

Q1. Find the slope of the line through $(-3,\ 2)$ and $(5,\ -6)$.

Q2. Find the inclination of a line whose slope is $-\sqrt{3}$.

Q3. The line through $(2,\ k)$ and $(4,\ 7)$ is parallel to the line through $(0,\ 1)$ and $(3,\ 7)$. Find k.

Q4. Write the equation of the line through $(3,\ -5)$ that is parallel to the x-axis.

Q5. Write the equation of the line with slope $\dfrac{2}{3}$ and y-intercept -4.

Q6. Find the intercepts that the line $4x - 3y + 24 = 0$ makes on the two axes.

Q7. Find k if the line $kx + 4y - 7 = 0$ is perpendicular to the line $3x - 2y + 5 = 0$.

Q8. Find the acute angle between the line $x - y = 4$ and the y-axis.

Q9. Find the distance of the point $(-2, 3)$ from the line $12x - 5y + 1 = 0$.

Q10. Find the distance between the parallel lines $5x - 12y + 7 = 0$ and $10x - 24y - 13 = 0$.

Q11. A line passes through $(-2, 5)$. Its x-intercept and y-intercept are equal in size but opposite in sign. Write its equation.

Q12. Fill in the blanks.

(a) The slope of a line perpendicular to $2x + 5y = 9$ is _____.

(b) The equation of the line parallel to the y-axis, 3 units to its left, is _____.

(c) The line $3x + 4y = 12$ meets the y-axis at the point _____.

(d) The equation of the line through the origin with inclination 60° is _____.

(e) The perpendicular distance of the origin from the line $3x - 4y = 10$ is _____.

(f) The slope of the line through (a, b) and (b, a), where $a \neq b$, is _____.

Q13. State True or False.

(a) The lines $2x - 3y = 5$ and $6y = 4x + 1$ are parallel.

(b) The slope of a vertical line is 0.

(c) The point $(2, -1)$ lies on the line $3x + 2y = 4$.

(d) The y-intercept of the line $2x - 5y + 10 = 0$ is -2.

(e) Two non-vertical lines with slopes m_1 and m_2 are perpendicular when $m_1 m_2 = -1$.

(f) The line $y = 3$ is parallel to the x-axis.

Q14. The acute angle between the lines $\sqrt{3}x - y + 2 = 0$ and $x - \sqrt{3}y - 1 = 0$ is

(a) 15° (b) 30° (c) 45° (d) 60°

Q15. The line $\dfrac{x}{3} + \dfrac{y}{4} = 1$ meets the x-axis at A and the y-axis at B. The length AB is

(a) 5 (b) 7 (c) 12 (d) 25

Q16. The equation of the line through $(-4, 1)$ that is perpendicular to the x-axis is

(a) $y = 1$ (b) $x = -4$ (c) $y = -4$ (d) $x = 1$

Q17. The line through $(2, 1)$ parallel to $3x + y = 5$ is

(a) $3x + y = 7$ (b) $x - 3y = -1$ (c) $3x - y = 5$ (d) $x + 3y = 5$

Q18. The lines $2x + y - 3 = 0$, $5x + ky - 3 = 0$ and $3x - y - 2 = 0$ pass through one point. The value of k is

(a) 2 (b) -2 (c) 3 (d) -3

Q19. The foot of the perpendicular from $(1, -2)$ to the line $x - y + 1 = 0$ is

(a) $(-1, 0)$ (b) $(0, 1)$ (c) $(1, 2)$ (d) $(-2, -1)$

Q20. A line through $P(3,\,-2)$ meets the axes so that P is the mid-point of the part between the axes. Its equation is

(a) $2x - 3y = 12$ (b) $3x - 2y = 13$ (c) $2x + 3y = 0$ (d) $x - y = 5$

Q21. The triangle formed by the lines $x = 0$, $y = 0$ and $x + y = 6$ is

(a) equilateral (b) right-angled and isosceles (c) obtuse-angled (d) scalene and acute-angled

Section B — Short answer

2 marks each

Q22. Find the equation of the line through $(2,\,-3)$ that makes an angle of 135° with the positive x-axis.

Q23. Find the equation of the line through $(-1,\,4)$ and $(3,\,-2)$. Write it in the form $ax + by + c = 0$.

Q24. Find the acute angle between two lines whose slopes are $2 - \sqrt{3}$ and $2 + \sqrt{3}$.

Q25. Find the points on the x-axis that are 4 units away from the line $4x + 3y = 12$.

Q26. Find the equations of the lines whose x-intercept and y-intercept have sum 5 and product 6.

Q27. Without finding any equation, show that the line through $(2,\,5)$ and $(-1,\,3)$ is perpendicular to the line through $(0,\,0)$ and $(2,\,-3)$.

Q28. The vertices of a triangle are $A(2,\,3)$, $B(-1,\,0)$ and $C(4,\,-2)$. Find the equation of the median through A.

Section C — Short answer

3 marks each

Q29. The vertices of a triangle are $A(1,\,4)$, $B(-2,\,1)$ and $C(5,\,-3)$. Find the equation of the altitude from A.

Q30. Find the equation of the perpendicular bisector of the segment that joins $(3,\,-1)$ and $(-5,\,5)$.

Q31. Find the image of the point $(1,\,2)$ in the line $2x - y + 5 = 0$. (The line is the perpendicular bisector of the point and its image.)

Q32. Find the point where the lines $2x + 3y = 13$ and $5x - y = 7$ meet. Then find the equation of the line through that point which is perpendicular to $3x - 4y + 1 = 0$.

Q33. Find the equations of the lines through $(3,\,-2)$ that make an angle of 60° with the line $\sqrt{3}x + y = 1$.

Q34. Find all values of k for which the point $(k,\,2)$ is 4 units away from the line $3x + 4y - 6 = 0$.

Q35. $A(2,\,1)$ and $C(6,\,5)$ are opposite vertices of a square $ABCD$. Find the equation of the diagonal BD, and the coordinates of B and D.

Section D — Long answer — 5 marks each

Q36. Find the equations of all lines that pass through $(2,\ 3)$ and are at a distance of 2 units from the origin. (Do not forget a line that has no slope.)

Q37. The vertices $A(-2,\ 3)$ and $B(2,\ 1)$ of a rectangle $ABCD$ lie on the line $x + 2y - 4 = 0$. The vertex C lies on the x-axis. Find the equations of the other three sides, and the coordinates of C and D.

Q38. A line meets the x-axis at A and the y-axis at B. The point $P(3,\ -4)$ divides AB internally in the ratio $AP : PB = 2 : 3$.

(a) Find the coordinates of A and B.

(b) Find the equation of the line.

(c) Find the length AB and the area of $\triangle OAB$, where O is the origin.

Q39. The vertices of a triangle are $A(0,\ 0)$, $B(6,\ 0)$ and $C(2,\ 4)$.

(a) Find the equations of the altitudes from A and from C, and hence the orthocentre.

(b) Find the equations of the perpendicular bisectors of AB and AC, and hence the circumcentre.

(c) Check that the circumcentre is equally far from A, B and C.

Section E — Case study — 4 marks

Q40. A town planner draws a map on a coordinate grid, where 1 unit $= 100$ m. The school is at the origin $O(0,\ 0)$. A straight main road runs along the line $3x + 4y = 24$. The town wants a footpath from the school to the road that is as short as possible.

(i) Find the points where the road crosses the x-axis and the y-axis. **(1 mark)**

(ii) Find the slope of the road. **(1 mark)**

(iii) Find the length of the shortest footpath from the school to the road, in metres. **(2 marks)**

OR

(iii) Find the equation of the footpath that meets the road at right angles, and the point where it meets the road. **(2 marks)**

— End of worksheet —
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